NOMSA MALABA

Junior Data Scientist | Python • SQL • Statistics • Machine Learning

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PROFILE

UCT BSc Applied Statistics and Psychology graduate with two years of Computer Science studies and practical experience in Python, SQL, machine learning and database-backed development. Currently a Software engineering intern working with Azure ML, Python and Django/PostgreSQL systems. Demonstrable knowledge of applied statistical modelling, data cleaning, reproducible preprocessing and model evaluation in a customer-churn project. Combines quantitative analysis, behavioural science and previous healthcare experience to support evidence-based decisions and meaningful health outcomes.

TECHNICAL SKILLS

Data science: Python, R, pandas, NumPy, scikit-learn, Jupyter, exploratory analysis, feature preprocessing, supervised learning, unsupervised learning

Statistics: Regression, hypothesis testing, ANOVA, experimental design, model evaluation, precision/recall, ROC-AUC

Data & engineering: SQL, PostgreSQL, SQLite, relational data, Django, REST APIs, Git, testing, documentation, Azure ML, Odoo, XML, JavaScript, HTML, CSS

PROFESSIONAL EXPERIENCE

Software Engineer | ZTQ Solutions | 2026 (May) –Present

- Contribute to Django/PostgreSQL applications, structured data workflows, REST APIs and role-based business functionality.
- Investigate and debug application/data issues; test changes and document findings in collaboration with developers and stakeholders.
- Translate requirements into practical technical solutions, reporting improvements and user-facing functionality.

Data Scientist & Automation Trainee | 2026 (Feb) – 2026 (April)

- Work with Python, Microsoft Azure and Azure Machine Learning on data-oriented and automation tasks.

SELECTED PROJECTS

Customer Churn Prediction | Python, pandas, scikit-learn, Jupyter

- Cleaned and analysed 7,000+ telecom customer records; explored behavioural and service patterns associated with churn.
- Built preprocessing/classification pipelines and compared Logistic Regression with Random Forest; selected model achieved ROC-AUC 0.836.
- Adjusted classification threshold from 0.50 to 0.40, increasing churn recall from 57% to 68%; translated results into retention recommendations and documented limitations.

github.com/Nomsa02/customer-churn-ml

Career Compass | Python, Django, PostgreSQL, SQL

- Developed a database-backed job tracking application with relational models, authentication, user-specific access control and CRUD workflows; used Git and PostgreSQL for deployment.

github.com/Nomsa02/Career-compass

EDUCATION & CREDENTIALS

University of Cape Town | BSc Applied Statistics & Psychology

Additional studies: Two years of Computer Science | Academic recognition: Dean's Merit List, Class Medal, Humanities Faculty Scholarship

Certification: ISC^2 Certified in Cybersecurity (CC) | Additional training: Microsoft Azure Data Science Associate; Power Automate / RPA (training only)

STA2020F (Applied Statistics): Applying statistical methods and modelling techniques to data Analysis of variance and experimental design; revision and extension of simple linear regression; multiple regression; logistic regression; model building; time series analysis; and non-parametric statistics. Analysis of data using R

STA3030F (Statistical Inference and Modelling): Computer simulation and data re-sampling techniques (bootstrap); Generalized linear models module extends linear regression models to models for non-normal response variables, including logistic regression. Machine learning: basic introduction to statistical learning paradigms, applications of regression and classification trees, and feedforward neural networks and backpropagation. Bayesian module: decision theory and Bayesian inference.